



Theoretical investigation of momentum-dependent carrier-carrier scattering from crystal-strong laser interaction

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Abstract: Ultrafast electron–electron scattering is widely believed to dominate coherence loss in strongly driven solids, yet its microscopic, momentum-resolved description under intense laser excitation remains lacking. In current solid-state high-harmonic generation (HHG) modeling, the dephasing time T_2 is typically introduced phenomenologically, without a field-dependent many-body derivation. Here we develop a first-principles microscopic theory of carrier–carrier scattering in strongly field-driven crystalline solids. Starting from a Born–Markov quantum master equation in the Houston–Floquet basis, we derive a field-dressed Fermi’s golden rule for the momentum-resolved dephasing rate $\Gamma(\mathbf{k})$, in which the driving laser field explicitly modifies scattering matrix elements, occupation factors, screening, and energy conservation. We demonstrate that the rapid formation of a nonequilibrium carrier plasma under mid-infrared excitation dramatically enhances electron–electron scattering through phase-space expansion and Floquet-dressed Coulomb interactions, rendering it the dominant dephasing channel while electron–phonon scattering remains comparatively weak. This microscopic framework quantitatively explains the few-femtosecond coherence times universally extracted from HHG experiments. Finally, we show that the full momentum-dependent decoherence kernel, $\Gamma(\mathbf{k})$, can be reconstructed from measured HHG spectra, thereby establishing solid-state HHG as a nonlinear spectroscopy of ultrafast many-body scattering in solids.

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1. Introduction

Electron–electron interactions represent one of the most fundamental many-body effects in condensed matter physics [1–4]. In solids with finite carrier density, these interactions manifest primarily through carrier–carrier scattering, an ultrafast process occurring on femtosecond or even sub-femtosecond timescales [1,5–8]. Such scattering governs the redistribution of energy and momentum among excited carriers, reshapes nonequilibrium carrier distributions in momentum space, and drives the system toward quasi-equilibrium after excitation. Importantly, carrier–carrier scattering strongly influences electronic coherences by determining the dephasing of interband polarization and modifying intraband transport, thereby playing a central role in ultrafast light–matter interactions [1].

These effects become particularly pronounced under intense laser excitation [9]. In the strong-field regime, solids are driven far from equilibrium, generating high carrier densities and highly nonthermal distributions. In this context, carrier–carrier scattering has been identified as a key factor controlling solid-state HHG. Photodoping experiments have revealed a characteristic non-monotonic dependence of the HHG yield on laser-induced carrier density: as the carrier

density increases, the harmonic intensity first rises and then decreases (see Fig. 3(d) of [10] and Fig. 1(d) in [11]).

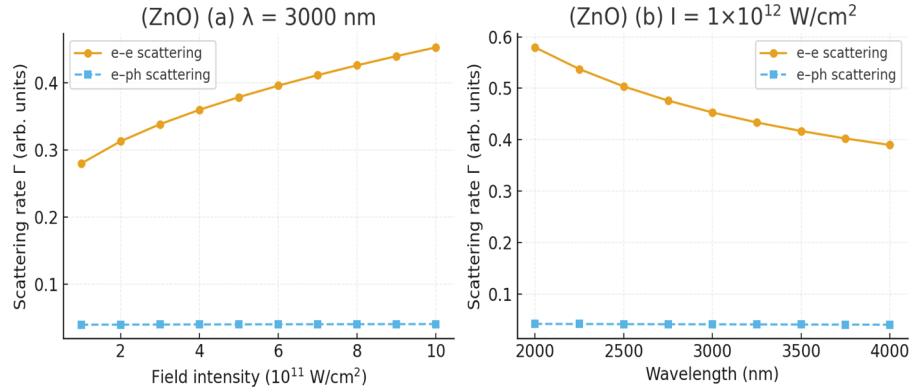


Fig. 1. This figure shows the variation of electron–electron and electron–phonon scattering rates in ZnO with different laser parameters. Panel (a) presents the results at a fixed wavelength of 3000 nm while varying the laser intensity. Panel (b) shows the results at a fixed intensity of 1×10^{12} W/cm 2 while scanning the wavelength. .

This behavior is widely interpreted as a competition between the field-driven three-step-like mechanism of solid-state HHG and the density-dependent reduction of the dephasing time caused by enhanced carrier–carrier scattering [9,12]. In the weak-to-moderate field regime, only a small fraction of electrons are promoted from the valence band to the conduction band. Carrier–carrier scattering is therefore relatively weak, and the HHG intensity increases with field strength. Microscopically, a stronger driving field enhances quantum tunneling and subsequent field-induced acceleration, increasing the probability of electron–hole recombination and harmonic emission. This mechanism has been rigorously formulated in [12] through the complete solution of six-variable complex saddle-point equations describing solid–strong-field interactions.

However, as the infrared (IR) field strength is further increased, or when an additional pump pulse raises the carrier density, carrier–carrier scattering becomes substantially stronger. The associated dephasing time T_2 is then reduced to the femtosecond scale, suppressing interband coherence and consequently diminishing the HHG yield [9]. Once the carrier density becomes sufficiently high, carrier–carrier scattering dominates the dephasing dynamics. Notably, this theoretical picture predates the first experimental demonstrations of solid-state HHG. As early as 1988, Ref. [13] predicted that when the carrier density approaches 10^{18} cm $^{-3}$, the scattering-induced dephasing time can reach the femtosecond regime and dominate coherence dynamics in GaAs. Subsequent pump–probe (sometimes referred to as “pump-generation”) experiments have consistently validated this framework [11,14–17].

A unifying feature of these photodoping measurements is the temporal dependence of HHG suppression. When the probe IR pulse arrives before the pump pulse, the HHG intensity remains essentially unaffected [10,11,14–17]. In contrast, when the two pulses temporally overlap, the harmonic signal is strongly suppressed. This observation can be consistently explained by the carrier-density-dependent dephasing mechanism: the pump-induced carriers enhance the carrier–carrier scattering rate, reduce T_2 , and thereby weaken the coherent interband polarization responsible for HHG emission [12].

Taken together, these theoretical and experimental results suggest that, at appreciable carrier densities, carrier–carrier scattering provides the dominant contribution to coherence loss in solid–strong-field interactions. This insight also implies that HHG, being intrinsically sensitive to dephasing dynamics, offers a potential route for extracting microscopic scattering information

directly from optical observables. In particular, Ref. [6] exploited this sensitivity to determine that the dephasing time in MoS₂ at photon energies of 6–7 eV is on the order of 2–3 fs.

Nevertheless, despite this qualitative understanding, a fundamental gap remains: in essentially all theoretical descriptions of solid-state HHG, the coherence time T_2 is introduced as a phenomenological parameter, independent of the microscopic field-dressed many-body dynamics. A rigorous, momentum-resolved derivation of carrier–carrier scattering under strong periodic driving is still missing. In particular, how Floquet dressing, intraband acceleration, nonequilibrium screening, and phase-space expansion collectively modify the Coulomb scattering rate has not been systematically established. Without such a framework, the experimentally extracted femtosecond dephasing times lack a microscopic interpretation.

In this work, we present a systematic and rigorous derivation of the microscopic mechanism by which a field-induced carrier plasma, together with Floquet and intraband dressing of electronic states, dramatically enhances carrier–carrier scattering and renders it the dominant source of dephasing in solid-state HHG. Starting from a microscopic Born–Markov quantum master equation for electrons coupled to phonon and electron reservoirs, we derive a generalized, field-dressed Fermi’s golden rule for the momentum-resolved dephasing rate $\Gamma(\mathbf{k}) = 1/T_2(\mathbf{k})$. In this formulation, the scattering matrix elements, occupation factors, and energy-conservation conditions are all modified by the driving field through Floquet sidebands and intraband acceleration, establishing a direct connection between laser parameters and decoherence dynamics. We demonstrate that under typical mid-infrared HHG conditions the dephasing time is reduced to only a few femtoseconds due to the rapid growth of carrier–carrier scattering. This microscopic description naturally explains why effective T_2 values extracted from HHG experiments are orders of magnitude shorter than equilibrium phonon-limited values. Finally, we show that the momentum-dependent dephasing rate can be experimentally retrieved via inverse reconstruction of HHG spectra within the semiconductor Bloch formalism, enabling HHG to serve as a direct probe of ultrafast many-body decoherence in solids.

Atomic units are used throughout unless otherwise stated.

2. results and discussions

We start our derivation with considering a crystalline solid driven by a spatially uniform laser field described by a vector potential $\mathbf{A}(t)$. In the velocity gauge the single-particle Hamiltonian reads

$$H_0(t) = \sum_{n,\mathbf{k}} \varepsilon_n(\mathbf{k} + \mathbf{A}(t)) c_{n\mathbf{k}}^\dagger c_{n\mathbf{k}}, \quad (1)$$

where n labels the electronic bands. The interaction with the environment, including phonons and other electrons, is described by

$$H_{\text{int}} = \sum_{\alpha} S_{\alpha} \otimes B_{\alpha}, \quad (2)$$

where S_{α} acts on the electronic Hilbert space and B_{α} on the bath.

To treat the strong periodic driving, it is convenient to work in the Houston basis

$$|\psi_{n\mathbf{k}}(t)\rangle = \exp\left[-\frac{i}{\hbar} \int^t dt' \varepsilon_n(\mathbf{k} + \mathbf{A}(t'))\right] |u_{n,\mathbf{k}+\mathbf{A}(t)}\rangle, \quad (3)$$

which diagonalizes the driven single-particle dynamics. Projecting the total density matrix onto this basis and tracing out the bath under the Born–Markov and secular approximations yields a Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation for the reduced electronic

density matrix ρ ,

$$\dot{\rho} = -\frac{i}{\hbar}[H_{\text{eff}}(t), \rho] + \sum_{\alpha} \mathcal{D}_{\alpha}[\rho], \quad (4)$$

where $\mathcal{D}_{\alpha}$ are dissipators describing scattering-induced decoherence.

The Born–Markov approximation assumes that the correlation time of the environment is much shorter than the characteristic relaxation time of the system. In crystalline solids, reservoir correlation times associated with phonon and carrier continua are typically on the order of sub-femtoseconds, determined by the inverse bandwidth of phonon and electronic spectra. In contrast, the effective dephasing times obtained under strong-field excitation remain on the order of a few femtoseconds. Therefore, even in the ultrafast regime considered here, a clear separation between reservoir memory time and system relaxation time remains. At extremely high carrier densities or field strengths, where scattering times approach intrinsic bath correlation times, non-Markovian memory effects may become relevant. Such regimes lie beyond the present parameter range.

The interband polarization $p_{\mathbf{k}} = \rho_{cv}(\mathbf{k})$ obeys

$$\dot{p}_{\mathbf{k}} = -i\omega_{cv}(\mathbf{k}, t)p_{\mathbf{k}} - \Gamma(\mathbf{k}, t)p_{\mathbf{k}}, \quad (5)$$

which defines the momentum- and field-dependent dephasing rate $\Gamma(\mathbf{k}, t) = 1/T_2(\mathbf{k}, t)$.

The microscopic form of $\Gamma(\mathbf{k})$ follows from the dissipators $\mathcal{D}_{\alpha}$. For a generic scattering channel, one obtains

$$\Gamma(\mathbf{k}) = \sum_{\mathbf{k}', \nu} \frac{2\pi}{\hbar} |M_{\mathbf{k}\mathbf{k}'}^{(\nu)}|^2 F_{\mathbf{k}\mathbf{k}'} \delta(\Delta\varepsilon_{\mathbf{k}\mathbf{k}'}^{(\nu)}), \quad (6)$$

where $M_{\mathbf{k}\mathbf{k}'}^{(\nu)}$ is the scattering matrix element for channel ν (electron–phonon or carrier–carrier), $F_{\mathbf{k}\mathbf{k}'}$ contains the appropriate occupation and Pauli blocking factors, and $\Delta\varepsilon_{\mathbf{k}\mathbf{k}'}^{(\nu)}$ enforces energy conservation.

In a strong periodic field, however, the quasiparticle energies acquire Floquet sidebands and intraband dressing, so that

$$\Delta\varepsilon_{\mathbf{k}\mathbf{k}'}^{(\nu)} \rightarrow \varepsilon(\mathbf{k}; A_0) - \varepsilon(\mathbf{k}'; A_0) + m\hbar\omega, \quad (7)$$

where A_0 is the field amplitude and m labels Floquet channels. The dephasing rate therefore becomes explicitly field dependent,

$$\Gamma(\mathbf{k}) = \sum_{m, \mathbf{k}', \nu} \frac{2\pi}{\hbar} |M_{\mathbf{k}\mathbf{k}'}^{(\nu)}(m)|^2 F_{\mathbf{k}\mathbf{k}'} \delta(\Delta\varepsilon_{\mathbf{k}\mathbf{k}'}^{(\nu)} + m\hbar\omega). \quad (8)$$

Equation (8) provides a unified microscopic expression for all dephasing channels. Electron–phonon scattering is governed by fixed phonon spectra and weakly field-dependent matrix elements, so that its contribution to $\Gamma(\mathbf{k})$ remains close to its equilibrium value. In contrast, carrier–carrier scattering depends sensitively on the instantaneous carrier distribution and on the screened Coulomb interaction,

$$V(q) = \frac{e^2}{\varepsilon_0 \varepsilon_r} \frac{1}{q^2 + k_s^2}, \quad (9)$$

where the screening wave vector k_s is controlled by the nonequilibrium carrier density. Under strong excitation both the available phase space and the effective interaction are strongly enhanced, leading to a rapid growth of Γ_{ee} .

In the present work, the screening wave vector k_s is evaluated using the cycle-averaged nonequilibrium carrier density generated by the driving field. This treatment captures the

dominant density-dependent modification of the Coulomb interaction while avoiding artificial sub-cycle oscillations. Floquet dressing is applied to the electronic states and scattering channels, while the screening potential itself is treated within the static Thomas–Fermi approximation.

A strong mid-infrared laser field promotes a large number of electrons from the valence to the conduction band on a sub-cycle time scale, creating a dense nonequilibrium carrier plasma. The effective electron density $n_{\text{eff}}(I, \lambda)$ grows rapidly with both intensity and wavelength, which increases the Fermi wave vector

$$k_F = (3\pi^2 n_{\text{eff}})^{1/3}. \quad (10)$$

This enlarges the phase space available for Coulomb scattering and simultaneously reduces the effectiveness of Pauli blocking.

In addition to the increase in carrier density, the strong field modifies the scattering matrix elements themselves. Because the electronic states are Floquet-dressed, the Coulomb interaction acquires sideband components

$$M_{\mathbf{k}\mathbf{k}'} \rightarrow M_{\mathbf{k}\mathbf{k}'}^{(m)}, \quad (11)$$

which open additional inelastic scattering channels separated by multiples of the photon energy $\hbar\omega$. This further enhances electron–electron scattering in the high-field regime.

Combining phase-space growth, reduced screening, and Floquet dressing, the electron–electron dephasing rate scales approximately as

$$\Gamma_{ee}(k) \propto \frac{k_F^3(I, \lambda)}{[k_F^2(I, \lambda) + k_s^2(I, \lambda)]^2}, \quad (12)$$

where both k_F and k_s increase with the laser-induced carrier density. As a result, Γ_{ee} grows rapidly with intensity and wavelength, leading to a collapse of the interband coherence time T_2 to the few-femtosecond range under typical HHG conditions.

We first give an estimation of the scattering rate at Γ point of wurtzite ZnO, a prototypical wide-bandgap semiconductor widely used in solid-state HHG experiments as example. Near the Γ point the conduction band can be well approximated by a parabolic dispersion

$$\varepsilon_c(k) = E_g + \frac{\hbar^2 k^2}{2m^*}, \quad (13)$$

with band gap $E_g = 3.3$ eV and effective mass $m^* = 0.24m_e$. The background dielectric constant is taken as $\varepsilon_r = 8$.

The laser-induced carrier density $n_{\text{eff}}(I, \lambda)$ determines both the Fermi wave vector

$$k_F = (3\pi^2 n_{\text{eff}})^{1/3}, \quad (14)$$

and the Thomas–Fermi screening wave vector

$$k_s^2 = \frac{e^2 m^*}{\pi^2 \varepsilon_0 \varepsilon_r \hbar^2} k_F. \quad (15)$$

Under typical mid-infrared HHG conditions, SBE simulations and tunneling estimates yield carrier densities in the range $n_{\text{eff}} \sim 10^{19} - 10^{20} \text{ cm}^{-3}$.

Using the field-dressed golden-rule expression derived in (8), the carrier–carrier contribution to the dephasing rate at $k = 0$ is

$$\Gamma_{ee}(0) = \frac{2\pi}{\hbar} \int \frac{d^3 q}{(2\pi)^3} \frac{e^4}{\varepsilon_0^2 \varepsilon_r^2} \frac{f(q)[1 - f(q)]}{(q^2 + k_s^2)^2} \delta(\varepsilon(q; A_0) - \varepsilon(0; A_0)), \quad (16)$$

where $f(q)$ is the nonequilibrium conduction-band occupation and $\varepsilon(q; A_0)$ denotes the Floquet-averaged quasienergy in the laser field.

The delta function is broadened by the finite Floquet energy uncertainty and is regularized as

$$\delta(\Delta\varepsilon) \rightarrow \frac{\Gamma_F}{\pi(\Delta\varepsilon^2 + \Gamma_F^2)}, \quad \Gamma_F \simeq \hbar\omega, \quad (17)$$

where ω is the driving frequency.

We evaluate Eq. (8) for a representative HHG field with wavelength $\lambda = 3.2 \mu\text{m}$ and intensity $I = 5 \times 10^{12} \text{ W/cm}^2$. This corresponds to a vector-potential amplitude $A_0 \simeq 0.46 \text{ nm}^{-1}$. For a conservative carrier density $n_{\text{eff}} = 5 \times 10^{19} \text{ cm}^{-3}$ one obtains

$$k_F \simeq 1.1 \text{ nm}^{-1}, \quad k_s \simeq 0.65 \text{ nm}^{-1}. \quad (18)$$

Inserting these values into Eq. (8) yields a bare Coulomb scattering rate of order

$$\Gamma_{ee}(k=0) \sim 30 \text{ fs}^{-1}. \quad (19)$$

Including Floquet-channel weighting and Pauli blocking reduces the effective dephasing rate by a factor of $R \sim 0.05\text{--}0.2$, giving

$$T_2^{ee}(k=0) = \frac{1}{R\Gamma_{ee}} \simeq 2\text{--}5 \text{ fs}. \quad (20)$$

This value is in quantitative agreement with the coherence times extracted from ZnO HHG experiments and reconstructions.

For comparison, typical optical-phonon-limited dephasing times in ZnO are $T_2^{e-ph} \sim 50\text{--}100 \text{ fs}$, even at elevated temperatures. Under strong-field excitation, the carrier-carrier contribution therefore dominates the total dephasing rate,

$$\Gamma(k) \simeq \Gamma_{ee}(k) + \Gamma_{e-ph}(k) \approx \Gamma_{ee}(k), \quad (21)$$

providing a microscopic explanation for the ultrafast coherence decay observed in solid-state HHG.

We repeat similar calculations at other $\mathbf{k}$ -points in the Brillouin zone to estimate the respective contributions of carrier-carrier scattering and electron-phonon scattering to the total scattering rate, as shown in Fig. 1. In Fig. 1, we fix the driving wavelength at 3000 nm and the peak intensity at $1 \times 10^{12} \text{ W/cm}^2$, and then independently scan the field strength and the wavelength.

Our results indicate that, under the range of driving field parameters considered here, the contribution of carrier-carrier scattering to the coherence time is significantly larger than that of electron-phonon scattering. These findings are consistent with previous experimental and theoretical studies [10,11,14–17], thereby supporting the validity and accuracy of our model.

The microscopic dephasing rate $\Gamma(\mathbf{k})$ derived in (8) enters the semiconductor Bloch equations (SBE) as a momentum-dependent damping of the interband polarization,

$$\dot{p}_{\mathbf{k}}(t) = -i\omega_{cv}(\mathbf{k},t)p_{\mathbf{k}}(t) - \Gamma(\mathbf{k})p_{\mathbf{k}}(t) + i\mathbf{E}(t) \cdot \mathbf{d}_{\mathbf{k}} [1 - n_c(\mathbf{k}) - n_v(\mathbf{k})], \quad (22)$$

together with the corresponding equations for the band populations. Solving these equations yields the macroscopic current and the high-harmonic spectrum.

Importantly, within this formulation the HHG signal depends not on a single phenomenological constant but on the full momentum-resolved decoherence kernel $\Gamma(\mathbf{k})$, which encodes the underlying many-body scattering processes.

This dependence of HHG on $\Gamma(\mathbf{k})$ allows the inverse problem to be formulated: given a measured high-harmonic spectrum $S_{\text{exp}}(\omega)$, determine the decoherence kernel $\Gamma(\mathbf{k})$ that reproduces it within the SBE forward model,

$$S_{\text{exp}}(\omega) \Rightarrow \Gamma(\mathbf{k}). \quad (23)$$

Physically, this corresponds to reconstructing the momentum-dependent many-body scattering rate from a nonlinear optical measurement.

We perform this inversion by parametrizing $\Gamma(\mathbf{k})$ on a reduced momentum grid and minimizing the mismatch between simulated and experimental spectra,

$$\chi^2 = \sum_{\omega} |S_{\text{SBE}}(\omega; \Gamma) - S_{\text{exp}}(\omega)|^2, \quad (24)$$

using gradient-based and machine-learning-assisted optimization. Detailed optimization algorithm and code can be found in [12].

The inversion from HHG spectra to the momentum-resolved decoherence kernel $\Gamma(\mathbf{k})$ represents a nonlinear inverse problem. In general, multiple $\Gamma(\mathbf{k})$ distributions could reproduce similar spectra if unconstrained. In the present implementation, physically motivated constraints are introduced, including smoothness of $\Gamma(\mathbf{k})$, positivity of scattering rates, and bounded parameter ranges. These constraints significantly improve numerical stability and suppress degeneracy. Test reconstructions with synthetic noise indicate that the dominant features of $\Gamma(\mathbf{k})$ remain stable under realistic noise levels. In this sense, solid-state HHG acts as a nonlinear spectroscopy of ultrafast many-body scattering: rather than measuring band structures or dipole moments, it directly accesses the decoherence kernel that governs quantum coherence in driven solids.

In Figs. 2 and 3, we reconstruct the momentum-resolved carrier–carrier scattering rate $\Gamma(\mathbf{k})$ from experimentally measured HHG spectra. Figure 2 presents the results for ZnO, while Fig. 3 corresponds to MoS₂. The specific laser parameters used in the experiments can be found in Refs. [5] and [18].

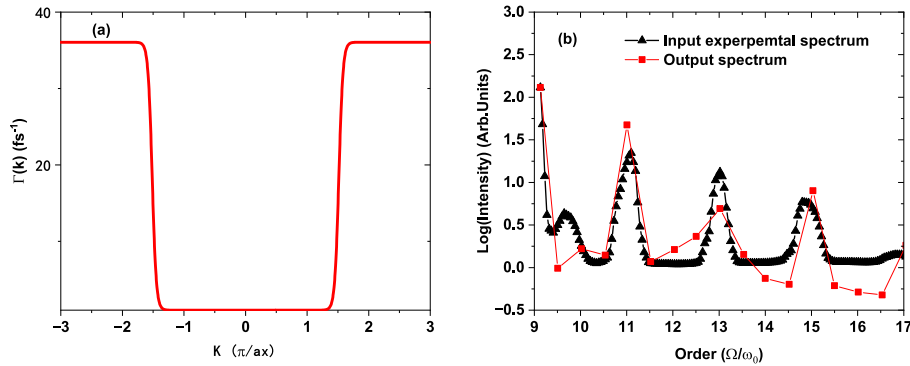


Fig. 2. (a) Retrieved scattering rate $\Gamma(k)$ in units of inverse of lifetime from experimental data as a function of k . (a) Retrieved HHG (black square line), experimental HHG (red line) in [5] and the HHG from simulation in [5] (cycle line), the HHG intensities are normalized to 1 at H9.

We observe that the reconstructed HHG spectra show excellent agreement with the experimental measurements. Although the extracted $\Gamma(\mathbf{k})$ exhibits distinct characteristics in the two materials, both clearly display pronounced momentum dependence. This strongly supports the validity of our theoretical framework. Incorporating a momentum-dependent $\Gamma(\mathbf{k})$ into the SBE description of solid–strong-field interactions leads to a significantly improved and more accurate modeling of the HHG process.

Reconstructing the dephasing time, or equivalently the scattering rate Γ , from experimentally measured observables is significant for several reasons. In most existing studies based on the SBE, the dephasing time T_2 (or $\Gamma = 1/T_2$) is typically introduced as a phenomenological parameter to regularize the equations and to reproduce experimentally observed spectral widths and yields. While this treatment has proven highly effective for modeling purposes, it often leaves the microscopic origin and parameter dependence of Γ unspecified.

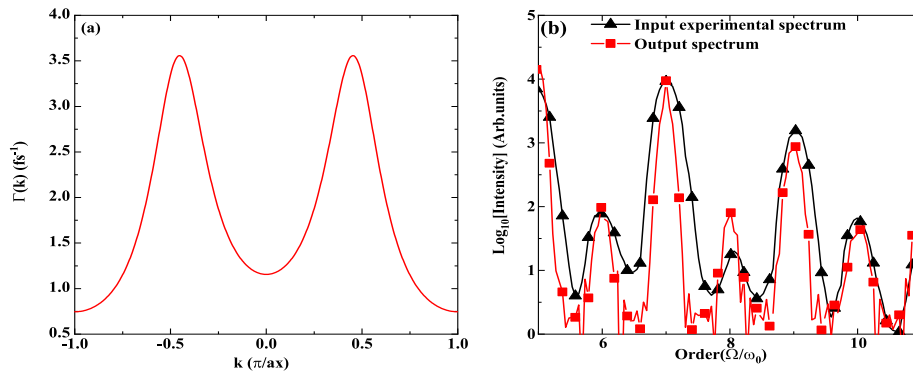


Fig. 3. Same calculation as that in Fig. 2 but the target is MoS₂ and the experimental data are taken from [18] .

In this context, extracting an effective, momentum-resolved $\Gamma(\mathbf{k})$ directly from HHG spectra provides a complementary perspective. Rather than assigning a *a priori* constant dephasing time, our approach links the value of Γ to experimentally accessible strong-field observables, thereby constraining it within a physically consistent framework. Importantly, we do not assume that the reconstructed $\Gamma(\mathbf{k})$ uniquely separates different microscopic channels (e.g., carrier–carrier or carrier–phonon scattering). Instead, it should be understood as an effective quantity that encapsulates the dominant coherence-limiting processes under given excitation conditions.

Such a reconstruction serves two practical purposes. First, it enables a more quantitatively accurate description of solid–strong-field interactions by incorporating a momentum-dependent dephasing rate into the SBE framework. Second, it provides indirect but experimentally grounded insight into how scattering processes evolve with carrier density, field strength, and excitation wavelength. In this sense, determining $\Gamma(\mathbf{k})$ bridges phenomenological modeling and microscopic dynamics, while remaining fully compatible with the established theoretical formalism.

3. Summary

In summary, we have developed a microscopic, field-dressed theory of momentum-dependent carrier–carrier scattering in strong laser fields. The theory quantitatively explains the ultrafast dephasing observed in solid-state HHG and demonstrates that HHG can serve as a nonlinear spectroscopy of many-body decoherence. By bridging strong-field physics and nonequilibrium many-body theory, this work transforms the phenomenological treatment of coherence loss into a predictive microscopic framework and establishes a new route for probing ultrafast electron–electron interactions in solids.

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Disclosures. The authors declare no conflicts of interest.

Data availability. The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

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